

Effects of Project-Based Learning on Student Engagement

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Abstract

This study investigated the effects of project-based learning (PBL) on student engagement in undergraduate computer science education.

1. Introduction

Project-based learning has been proposed as an effective pedagogy for STEM education. Previous studies suggest that PBL can increase student engagement and motivation.

2. Methods

We conducted a quasi-experimental study with two cohorts of N=120 undergraduate students. The treatment group (n=60) participated in a PBL course, while the control group (n=60) participated in a traditional lecture-based course.

3. Results

Students in the PBL group reported significantly higher engagement scores ($M=4.2$, $SD=0.6$) than the control group ($M=3.4$, $SD=0.5$), $t(118)=4.5$, $p<0.001$.

4. Discussion

The results indicate that project-based learning substantially improves student engagement compared to traditional lectures. These findings have implications for the design of undergraduate STEM courses.

5. Limitations

The study was limited to one university and one discipline. Generalization to other contexts requires further research. Self-reported engagement scores may be subject to bias.

6. Conclusion

Project-based learning shows promise for increasing engagement in undergraduate STEM education. Further research is needed to explore the long-term effects and implementation challenges.